NATIONAL RESEARCH FOUNDATION

NEW ROADMAP TO RESEARCH

By Amarnath K. Menon

More than four years after Prime Minister Narendra Modi pledged to launch an apex National Research Foundation (NRF), the plan is coming to fruition with a bill being readied for debate and adoption in the monsoon session of Parliament. At the inauguration of the 106th session of the Indian Science Congress in 2019, Modi declared that such a far-reaching initiative was imperative in line with the vision of restoring India's past glory as 'Vishwa Guru'. The goal is to make India a global leader in research and innovation. The delay is attributed to the disruption caused by the Covid-19 pandemic.

On approval by Parliament, the NRF will be set up later this year to provide "high-level strategic direction" to scientific research as recommended by the National Education Policy (NEP), 2020, with an investment of Rs 50,000 crore between 2023 and 2028.

The NRF will be a unique public-private partnership (PPP) entity, for which Rs 36,000 crore of the research funding for the first five years is to come from private partners, mostly industry. It will seed, grow and promote Research and Development (R&D) and foster a culture of research and innovation throughout India's universities, colleges, research institutions etc. "The NRF is to ensure scientific research is funded equitably and greater private participation is forthcoming," explains Jitendra Singh, the Union minister of state for science and technology. "We have also found that, so far, scientific research is conducted in silos—in government departments and laboratories, central and state universities—with no uniformity in infrastructure, such as the kind that th Indian Institutes of Technology have. There has also been a lack of uniformity in the funding of scientific research. These issues have been addressed in the bill."

While the premier research institutes get the bulk of the research funding, state universities barely get 10 per cent. The NRF will correct this. "This is welcome as the efforts made by higher education institutions to enhance the research ecosystem on their campuses will get an impetus," says UGC chairman M. Jagadesh Kumar. The Department of Science and Technology (DST) will continue as the main source of funds for several autonomous research bodies.

With the government pitching in with Rs 14,000 crore over five years, the hope is that the flow of private research grants—expected to be two and a half times that sum in the same period—will be smooth with the NRF in place.

ON APPROVAL BY PARLIAMENT, THE NRF WILL BE SET UP LATER THIS YEAR TO PROVIDE 'HIGH-LEVEL STRATEGIC DIRECTION' TO SCIENTIFIC RESEARCH

INNOVATION IN INDIA

#40

POSITION AMONG 132 NATIONS IN THE WIPO GLOBAL INNOVATION INDEX 2022

#46

AMONG 158 COUNTRIES IN THE UNCTAD TECHNOLOGY AND INNOVATION REPORT 2023



MANDAR DEODHAR

A study by the NITI Aayog and the Institute for Competitiveness shows that India's R&D spend is among the lowest in the world. The country currently spends 0.7 per cent of its GDP on R&D. The gross expenditure declined from 0.84 per cent in 2008 to about 0.69 per cent in 2018, the last year for which figures are available. This is despite India's spending on R&D consistently increasing in the past 10 years, with the country's estimated expenditure being Rs 1.24 lakh crore in 2018-19. In comparison, the US spent 2.8 per cent, China 2.1 per cent and Israel 4.9 per cent of their GDP on R&D. South Korea's spending is nearing 5 per cent now, of which four-fifths is from the private sector.

India has 366 R&D personnel per million population, compared to 1,366 in Brazil, 2,366 in China and 6,995 in Germany, according to a UNESCO report. Iceland (10,073) has the largest number per million. According to the